

Industry-Oriented Pandas Problem Statements with Mini Project Extensions

Note: Mini project extensions part is optional. Advanced learners are encouraged to carryout mini project.

1. E-Commerce Sales Analytics

Title

Product Sales Analysis Using Pandas

Problem Statement

An e-commerce company maintains product sales records containing:

- ProductID
- SalesAmount
- Category

Create a DataFrame for 20 products and perform the following operations:

1. Select only ProductID and SalesAmount columns.
2. Filter products with SalesAmount greater than ₹50,000.
3. Add a new column Region with the default value "South India".
4. Sort products by SalesAmount in descending order.
5. Calculate the average SalesAmount of all products.

Mini Project Extension

Develop a **Sales Analytics Dashboard** that:

- Displays top-selling products.
- Shows category-wise sales performance.
- Identifies low-performing products.
- Calculates monthly and quarterly sales trends.
- Generates automated sales reports.
- Visualizes sales using bar charts and pie charts.

Learning Outcome

Students learn product performance analysis and business intelligence reporting.

2. Banking Analytics

Title

Loan Application Analysis Using Pandas

Problem Statement

A bank maintains loan records consisting of:

- LoanID
- LoanAmount
- Branch

Create a DataFrame and perform:

1. Display LoanID and LoanAmount.
2. Filter loans greater than ₹10,00,000.
3. Add LoanType with default value "Personal Loan".
4. Sort records based on LoanAmount.
5. Calculate the average loan amount.

Mini Project Extension

Develop a **Loan Analytics and Risk Monitoring System** that:

- Categorizes loans into risk groups.
- Identifies high-value loans.
- Generates branch-wise loan statistics.
- Tracks loan approval trends.
- Creates dashboards and management reports.

Learning Outcome

Students apply Pandas for banking and financial analytics.

3. Healthcare Analytics

Title

Patient Health Monitoring Analysis Using Pandas

Problem Statement

A hospital stores patient information containing:

- PatientID
- BloodPressure
- Department

Perform the following:

1. Display PatientID and BloodPressure.
2. Filter patients with BloodPressure greater than 140.
3. Add Hospital column with default value "Apollo Hospital".
4. Sort the records based on BloodPressure.
5. Calculate average BloodPressure.

Mini Project Extension

Develop a **Hospital Analytics Dashboard** that:

- Detects high-risk patients.
- Generates department-wise health statistics.
- Identifies patients requiring immediate attention.
- Monitors patient health trends.
- Visualizes healthcare reports.

Learning Outcome

Students understand healthcare data analysis and patient monitoring.

4. Agriculture Analytics

Title

Crop Yield Analysis Using Pandas

Problem Statement

An agricultural organization maintains records containing:

- FarmID
- Yield
- CropType

Perform:

1. Display FarmID and Yield.
2. Filter farms with Yield greater than 5000 kg.
3. Add Season with default value "Rabi".
4. Sort records by Yield.
5. Calculate average crop yield.

Mini Project Extension

Develop a **Smart Agriculture Analytics System** that:

- Monitors farm productivity.
- Compares crop performance.
- Generates region-wise yield reports.
- Identifies high-performing farms.
- Predicts seasonal productivity trends.

Learning Outcome

Students learn agricultural data analytics using Pandas.

5. Cybersecurity Analytics

Title

Security Incident Analysis Using Pandas

Problem Statement

A cybersecurity company maintains incident records containing:

- IncidentID
- SeverityScore
- AttackType

Perform:

1. Select IncidentID and SeverityScore.
2. Filter incidents with SeverityScore greater than 8.
3. Add Status with default value "Open".
4. Sort incidents by SeverityScore.
5. Calculate average SeverityScore.

Mini Project Extension

Develop a **Security Operations Center (SOC) Dashboard** that:

- Tracks critical incidents.
- Categorizes threats by severity.
- Generates threat intelligence reports.
- Monitors attack trends.
- Provides incident response metrics.

Learning Outcome

Students apply data analytics to cybersecurity operations.

6. Artificial Intelligence Analytics

Title

AI Model Performance Analysis Using Pandas

Problem Statement

An AI company maintains model performance records containing:

- ModelID
- Accuracy
- Algorithm

Perform:

1. Select ModelID and Accuracy.
2. Filter models with Accuracy greater than 90%.
3. Add Status with default value "Production Ready".
4. Sort models by Accuracy.
5. Calculate average model accuracy.

Mini Project Extension

Develop an **AI Model Evaluation Dashboard** that:

- Ranks machine learning models.
- Compares algorithm performance.
- Tracks accuracy improvements.
- Generates automated evaluation reports.
- Visualizes model comparison results.

Learning Outcome

Students gain experience in machine learning model analytics.

7. Smart City IoT Analytics

Title

Sensor Monitoring Analysis Using Pandas

Problem Statement

A Smart City administration stores sensor records containing:

- SensorID
- Temperature
- Zone

Perform:

1. Display SensorID and Temperature.
2. Filter Temperature values above 40°C.
3. Add Project with default value "Smart City Mission".
4. Sort by Temperature.
5. Calculate average temperature.

Mini Project Extension

Develop a **Smart City Monitoring Dashboard** that:

- Monitors environmental conditions.
- Generates zone-wise statistics.
- Detects abnormal sensor readings.
- Generates automated alerts.
- Visualizes real-time temperature trends.

Learning Outcome

Students understand IoT data analytics and smart city applications.

8. Financial Market Analytics

Title

Stock Market Performance Analysis Using Pandas

Problem Statement

Create a DataFrame containing:

- StockSymbol
- ClosingPrice
- Sector

Perform:

1. Display StockSymbol and ClosingPrice.
2. Filter stocks with ClosingPrice greater than ₹1000.
3. Add Exchange with default value "NSE".
4. Sort stocks based on ClosingPrice.
5. Calculate average closing price.

Mini Project Extension

Develop a **Stock Market Analytics Dashboard** that:

- Tracks top-performing stocks.
- Compares sector performance.
- Generates investment insights.
- Monitors price trends.
- Produces daily market reports.

Learning Outcome

Students gain practical experience in financial data analytics.

9. Quantum Computing Research Analytics

Title

Quantum Experiment Performance Analysis Using Pandas

Problem Statement

A quantum computing laboratory stores experiment records containing:

- ExperimentID
- FidelityScore
- QuantumAlgorithm

Perform:

1. Display ExperimentID and FidelityScore.
2. Filter experiments with FidelityScore greater than 0.90.
3. Add Lab with default value "Quantum Research Center".
4. Sort experiments based on FidelityScore.
5. Calculate average FidelityScore.

Mini Project Extension

Develop a **Quantum Research Analytics Dashboard** that:

- Compares quantum algorithms.
- Monitors experiment success rates.
- Tracks fidelity improvements.
- Generates laboratory performance reports.
- Visualizes quantum experiment outcomes.

Learning Outcome

Students apply Pandas to emerging quantum computing research.

10. Cloud Computing Analytics (Advanced)

Title

Cloud Resource Utilization Analytics Using Pandas

Problem Statement

A cloud service provider maintains server utilization data containing:

- ServerID
- CPU_Utilization (%)
- DataCenter

Perform:

1. Display ServerID and CPU_Utilization.
2. Filter servers with utilization greater than 80%.
3. Add ServiceTier with default value "Premium".
4. Sort servers based on utilization.
5. Calculate average utilization.
6. Identify top utilized servers.

Mini Project Extension

Develop a **Cloud Infrastructure Monitoring Dashboard** that:

- Detects overloaded servers.
- Identifies underutilized resources.
- Generates DataCenter-wise utilization reports.
- Provides load-balancing recommendations.
- Monitors resource utilization in real time.
- Visualizes infrastructure health using charts.

Learning Outcome

Students learn cloud resource monitoring, infrastructure analytics, and operational decision support.
